

DIP-SMC-PSO:
Technical Progress
Memo

Model choices, algorithm design, and
experimental results

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Program: M.Sc. Program

February 2026 — Status: Research complete,
thesis 90%

Executive Summary

Problem. Stabilise a double-inverted pendulum (DIP) under model uncertainty and external disturbances using a single actuator ($u \in [-150, 150]$ N).

Approach. Four SMC variants (Classical, Super-Twisting, Adaptive, Hybrid Adaptive) PSO-tuned across 1,300+ simulations: 400 Monte Carlo runs, 15-scenario robust PSO, boundary layer study, and parametric uncertainty analysis.

Top result. STA-SMC: **1.82 s** settling time (−16% vs. Classical), **2.3%** overshoot, and **74%** chattering reduction (FFT metric). All three quantified improvements are statistically significant ($p \leq 0.038$; Cohen $d \geq 2.1$). MT-8 disturbance logs are retained as qualitative stress-test evidence only, because all controller-scenario entries are marked non-converged.

Current completion priorities. (1) Finalize STA gain-order consistency package for archival rigor (see submission checklist in Appendix B). (2) Keep Hybrid Adaptive outside MT-5 statistical ranking and retain it for methodological completeness only.

Thesis status. 90/100 score. January 2026 completion phases slipped due to ongoing research obligations. Revised target: **April 2026** (see updated timeline in Section 8.3).

Contributions

- **C1.** First systematic PSO-tuned boundary layer adaptation study on a 2-link DIP; includes honest correction of the 66.5% $\rightarrow$ 3.7% chattering reduction claim.
- **C2.** Robust multi-scenario PSO with worst-case fitness reduces the generalisation gap from 50.4 $\times$ to 7.5 $\times$ on challenging (± 0.30 rad) initial-condition distributions.
- **C3.** Simultaneous Lyapunov proofs (9 theorems, 5 lemmas) for all four SMC variants on the same DIP system, enabling direct stability-type comparison.

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1 System Model: Equations and Parameter Choices

1.1 State Vector and Coordinate Definition

- **DIP state vector:**

$$\mathbf{x} = [x, \theta_1, \theta_2, \dot{x}, \dot{\theta}_1, \dot{\theta}_2]^\top \in \mathbb{R}^6 \quad (1)$$

- **Coordinate convention:**

$$\mathbf{q} = [x, \theta_1, \theta_2]^\top, \quad \dot{\mathbf{q}} = [\dot{x}, \dot{\theta}_1, \dot{\theta}_2]^\top \quad (2)$$

Units: x [m], θ_i [rad], $\dot{x}$ [m/s], $\dot{\theta}_i$ [rad/s]. Sign convention: $+x$ rightward, $+\theta_i$ counterclockwise from upright.

- **Target equilibrium and error:**

$$\mathbf{x}^* = [0, 0, 0, 0, 0, 0]^\top, \quad \mathbf{e} = \mathbf{x} - \mathbf{x}^* \quad (3)$$

Tracking case: shift only the first component of $\mathbf{x}^*$.

- **Measured output:**

$$\mathbf{y} = [x, \theta_1, \theta_2, \dot{x}, \dot{\theta}_1, \dot{\theta}_2]^\top \quad (4)$$

Feedback source: full-state in simulation; encoder/estimator pipeline in HIL. Safety limits: benchmark configuration files.

1.2 Equations of Motion

- **Full nonlinear dynamics:**

$$\mathbf{M}(\mathbf{q}) \ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) \dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) + \mathbf{F}_{\text{fric}} = \mathbf{B} \mathbf{u} \quad (5)$$

with $\mathbf{q} = [x, \theta_1, \theta_2]^\top$, $\mathbf{B} = [1, 0, 0]^\top$, and actuator bounds $u \in [-150, 150]$ N.

- **Modeling assumptions:** planar rigid-body dynamics, ideal revolute joints, single-input bounded actuation, viscous + Coulomb friction, and disturbance injection through benchmark profiles (MT-8/LT-6).
- **Equivalent first-order form for control design:**

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) + \mathbf{g}(\mathbf{x}) u + \mathbf{d}(t) \quad (6)$$

Disturbance term: $\mathbf{d}(t) = 0$ (nominal), $\mathbf{d}(t) \neq 0$ (rejection tests).

- **Structural properties used in proofs:** Properties: $\mathbf{M}(\mathbf{q})$ symmetric positive definite; $\dot{\mathbf{M}}(\mathbf{q}) - 2\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ skew-symmetric on admissible configurations.
- **Model partition in this workflow:**
 - Full model: final benchmark reporting.
 - Simplified model: PSO loops (reduced coupling, no gyroscopic term).

Inertia matrix $\mathbf{M}(\mathbf{q})$ (3×3 , symmetric, configuration-dependent):

- Compact notation: $c_1 = \cos \theta_1$, $c_2 = \cos \theta_2$, $c_{12} = \cos(\theta_1 - \theta_2)$.

$$\mathbf{M}(\mathbf{q}) = \begin{bmatrix} M + m_1 + m_2 & a_{12} & a_{13} \\ a_{12} & a_{22} & a_{23} \\ a_{13} & a_{23} & a_{33} \end{bmatrix} \quad (7)$$

- Matrix entries:

$$\begin{aligned} a_{12} &= (m_1 l_{c1} + m_2 l_1) c_1 + m_2 l_{c2} c_2 \\ a_{13} &= m_2 l_{c2} c_2 \\ a_{22} &= m_1 l_{c1}^2 + m_2 (l_1^2 + l_{c2}^2) + I_1 + I_2 + 2m_2 l_1 l_{c2} c_{12} \\ a_{23} &= m_2 l_{c2}^2 + I_2 + m_2 l_1 l_{c2} c_{12} \\ a_{33} &= m_2 l_{c2}^2 + I_2 \end{aligned} \quad (8)$$

- Dominant coupling: a_{22} term $2m_2 l_1 l_{c2} c_{12}$; grows with large relative-link deflection.

Coriolis/centrifugal matrix $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ (non-zero entries):

- Computed from Christoffel symbols of (7); zero entries omitted.

$$\begin{aligned} C_{01} &= -(m_1 l_{c1} + m_2 l_1) \sin \theta_1 \dot{\theta}_1 - m_2 l_{c2} \sin \theta_2 \dot{\theta}_2 \\ C_{02} &= -m_2 l_{c2} \sin \theta_2 \dot{\theta}_2 \\ C_{11} &= C_{12} = -m_2 l_1 l_{c2} \sin(\theta_1 - \theta_2) \dot{\theta}_2 \\ C_{21} &= +m_2 l_1 l_{c2} \sin(\theta_1 - \theta_2) \dot{\theta}_1 \end{aligned} \quad (9)$$

- Indices $(0, 1, 2)$ correspond to (x, θ_1, θ_2) .
- Gyroscopic augmentation: $g_c(\dot{\theta}_1 + \dot{\theta}_2)$ added to C_{01} and $-C_{10}$; coefficient 0.01 (config-enabled).

Gravity and friction model:

$$\begin{aligned} \mathbf{G}(\mathbf{q}) &= \begin{bmatrix} 0 \\ G_2 \\ G_3 \end{bmatrix} \\ G_2 &= -(m_1 l_{c1} + m_2 l_1) g \sin \theta_1 \\ G_3 &= -m_2 l_{c2} g \sin \theta_2 \end{aligned} \quad (10)$$

- $G_1 = 0$ because gravity is perpendicular to cart translation.

$$\mathbf{F}_{\text{fric},i} = -b_i \dot{q}_i - \mu_i \text{sign}(\dot{q}_i) \mathbb{1}_{|\dot{q}_i| > 10^{-6}} \quad (11)$$

- Friction scope: all DOFs $i \in \{x, \theta_1, \theta_2\}$; dead zone $|\dot{q}_i| > 10^{-6}$ avoids sign discontinuity at rest.
- Viscous coefficients (0.2, 0.005, 0.004); Coulomb coefficients from config, Monte Carlo variation $\pm 10\%$.
- Simulation model: full viscous + Coulomb form (11); PSO speedup uses reduced coupling (0.7–0.8) and no gyroscopic term.
- Linearization: numerical $(\mathbf{A}, \mathbf{B})$ at $\mathbf{q}^* = \mathbf{0}, \dot{\mathbf{q}}^* = \mathbf{0}$ with $\varepsilon = 10^{-8}$; $\Delta t = 0.01$ s, $u \in [-150, 150]$ N.

1.3 Physical Parameters

Parameter	Symbol	Value
Cart mass	M	1.5 kg
Link 1 mass	m_1	0.2 kg
Link 2 mass	m_2	0.15 kg
Link 1 length	l_1	0.4 m
Link 2 length	l_2	0.3 m
Link 1 COM distance	l_{c1}	0.2 m
Link 2 COM distance	l_{c2}	0.15 m
Link 1 inertia	I_1	0.0081 kg·m ²
Link 2 inertia	I_2	0.0034 kg·m ²
Gravity	g	9.81 m/s ²
Cart friction	b_c	0.2 N·s/m
Joint 1 friction	b_1	0.005 N·m·s/rad
Joint 2 friction	b_2	0.004 N·m·s/rad
Max control force	$u_{\max}$	150 N
Control period	Δt	0.01 s (100 Hz)

1.4 Model Selection: Simplified vs. Full Non-linear

Model	Use in this project	Rationale
Simplified (linearized around upright)	PSO optimization (all controllers)	10–50× faster per simulation step. PSO requires thousands of evaluations; full model would be prohibitive.
Full nonlinear (5)	All published benchmark results	Complete Lagrangian, all coupling terms, Coriolis, centrifugal, gyroscopic, full friction. Used for MT-5, MT-7, MT-8, LT-6.
Low-rank (SVD approximation)	HIL mode (hardware-in-the-loop)	Reduced singular-value truncation preserves stability with lower real-time compute load.

- **Decision basis:** full nonlinear PSO ($40 \times 200 \times 10$ s) exceeds 8 h/controller.
- **Optimization runtime:** minutes on simplified model.
- **Reporting model:** full nonlinear only.

Simplification summary (Section 1.4, PSO only).

- **Baseline kept:** same state and actuation structure as (5) and (6), with unchanged bounds $u \in [-150, 150]$ N and $\Delta t = 0.01$ s.
- **Linearization step:** numerical $(\mathbf{A}, \mathbf{B})$ around upright equilibrium $(\mathbf{q}^*, \dot{\mathbf{q}}^*) = (\mathbf{0}, \mathbf{0})$ using $\varepsilon = 10^{-8}$.
- **Term reduction:** coupling terms scaled down (0.7–0.8 range) and gyroscopic augmentation disabled during PSO loops.
- **Scope guard:** all final figures/tables use the full nonlinear model; simplification is used only to reduce PSO search runtime (about 10–50×).

2 Sliding Mode Control: Surface Design and Control Laws

2.1 Sliding Surface

- **Compact state form:**

$$\sigma = \lambda_1(\theta_1 - \theta_1^*) + \lambda_2(\theta_2 - \theta_2^*) + k_1(\dot{\theta}_1 - \dot{\theta}_1^*) + k_2(\dot{\theta}_2 - \dot{\theta}_2^*) \quad (12)$$

- **Equivalent error form** ($e_i = \theta_i - \theta_i^*$, $\theta_i^* = 0$):

$$\sigma = \lambda_1 e_1 + \lambda_2 e_2 + k_1 \dot{e}_1 + k_2 \dot{e}_2 \quad (13)$$

- **Sliding dynamics, stability, and tuning:**

$$\sigma = 0 \Rightarrow \dot{e}_i = -\frac{\lambda_i}{k_i} e_i, \quad \lambda_i > 0, k_i > 0, i \in \{1, 2\} \quad (14)$$

Time constants: $\tau_i = k_i/\lambda_i$ (direct tuning meaning). MT-8 robust Classical example: $(\lambda_1, \lambda_2, k_1, k_2) = (5.51, 3.49, 23.07, 12.85)$ gives $\tau_1 \approx 4.19$ s and $\tau_2 \approx 3.68$ s.

- **Units and scaling:** Signal units: θ_i [rad], $\dot{\theta}_i$ [rad/s]. Scaling policy: no state normalization before PSO; scaling handled by gain bounds and cost normalization.

- **Relative degree, robustness, and practical limitation:**

$$\dot{\sigma} = \lambda_1 \dot{\theta}_1 + \lambda_2 \dot{\theta}_2 + \mathbf{L}\mathbf{M}^{-1}(\mathbf{B}\mathbf{u} - \mathbf{F}) + \mathbf{L}\mathbf{M}^{-1}\mathbf{B}d(t), \quad \mathbf{L} = [0, k_1, k_2], \quad \mathbf{F} = \mathbf{C}\dot{\mathbf{q}} + \mathbf{G} + \mathbf{F}_{\text{fric}}. \quad (15)$$

Reachability: select $K > \bar{d} + \eta$ when $|d(t)| \leq \bar{d}$. Limitation: surface excludes cart x ; mitigated by saturation $u \in [-150, 150]$ N and optional outer-loop cart reference shaping.

2.2 Controller Variants: Control Laws

- **Unified template (all variants):**

$$u = u_{\text{eq}} + u_{\text{sw}} + u_{\text{aux}} \quad (16)$$

Mapping: variant-specific choices of $(u_{\text{eq}}, u_{\text{sw}}, u_{\text{aux}})$.

2.2.1 Classical SMC: Saturation Function and Equivalent Control

- **Saturation function:**

$$\text{sat}\left(\frac{\sigma}{\varepsilon}\right) = \begin{cases} \sigma/\varepsilon & |\sigma| \leq \varepsilon \\ \text{sign}(\sigma) & |\sigma| > \varepsilon \end{cases} \quad (17)$$

- **Control law mapping to (16):**

$$u = \underbrace{u_{\text{eq}}}_{\text{equivalent}} + \underbrace{K \text{sat}\left(\frac{\sigma}{\varepsilon}\right)}_{\text{switching}} + \underbrace{-k_d \dot{\sigma}}_{\text{auxiliary damping}} \quad (18)$$

- **Equivalent control term (from $\dot{\sigma} = 0$):**

$$u_{\text{eq}} = -\frac{\mathbf{L}\mathbf{M}^{-1}\mathbf{F}}{\mathbf{L}\mathbf{M}^{-1}\mathbf{B}}, \quad \mathbf{F} = \mathbf{C}\dot{\mathbf{q}} + \mathbf{G} + \mathbf{F}_{\text{fric}} \quad (19)$$

- **Discrete-time implementation note** ($T_s = 0.01$ s):

$$\dot{\sigma}[k] = \beta_f \dot{\sigma}[k-1] + (1 - \beta_f) \frac{\sigma[k] - \sigma[k-1]}{T_s} \quad (20)$$

Sequence: compute u_{raw} via (18), then clamp $u[k] = \text{sat}_{[-150, 150]}(u_{\text{raw}}[k])$.

2.2.2 Super-Twisting Algorithm (STA-SMC)

- **Control decomposition:**

$$u = u_1 + u_2 \quad (21)$$

$$u_1 = -K_1 \sqrt{|\sigma| + \delta} \text{sign}(\sigma) \quad (\text{continuous})$$

$$u_2 = -K_2 \int_0^t \text{sign}(\sigma(\tau)) d\tau \quad (\text{integral memory})$$

Regularization: $\delta = 10^{-10}$ near $\sigma = 0$.

- **Integral anti-windup (implementation):**

$$u_2[k] = \text{clip}(u_2[k-1] - K_2 T_s \text{sign}(\sigma[k]), u_{2,\min}, u_{2,\max}) \quad (22)$$

- **Gain condition and consistency note:** Required condition: $K_1 > K_2 > 0$. Flagged record: $(K_1, K_2) = (2.02, 6.67)$ violates condition; needs re-tuning.

2.2.3 Adaptive SMC

- **Control law:**

$$u = -K(t) \text{sat}\left(\frac{\sigma}{\varepsilon}\right) \quad (23)$$

- **Adaptive gain update with projection and dead zone:**

$$\dot{K}(t) = \Pi_{[K_{\min}, K_{\max}]}(\gamma|\sigma| - \delta_{\text{leak}}K(t)), \quad |\sigma| \geq d_z \quad (24)$$

Dead-zone rule: $\dot{K}(t) = 0$ for $|\sigma| < d_z$.

- **Practical settings:** Bounds: $K(t) \in [0.1, 100]$; dead zone: $d_z = 0.05$; leak: $\delta_{\text{leak}} = 0.01$. Effect: prevents gain drift and noise-driven amplification.

2.2.4 Hybrid Adaptive STA-SMC

- **Control law:**

$$u = -k_1(t)|\sigma|^{0.5} \text{sign}(\sigma) - k_2(t) \int \text{sign}(\sigma) dt \quad (25)$$

Gain rates: $\gamma_1 = 0.1$, $\gamma_2 = 0.05$; rate limiter: 5.0/s.

Limitation (Hybrid Adaptive STA-SMC): Observed instability at large disturbances: up to 666.9° overshoot and 176% chattering increase. Status: retained for completeness; **not recommended for deployment**.

Hybrid failure diagnosis (MT-5/MT-8 evidence): Energy = 10^6 and chattering = 0 in MT-5 are sentinel values: the PSO instability penalty ($P = 10^6$) triggered on every run. The simulation did not abort; the controller was flagged as internally unstable.

The root cause is mode-switching instability: adaptive rates $\gamma_1=0.1$, $\gamma_2=0.05$ combined with a 5.0/s rate limiter are too aggressive for the rapid IC variations in the MT-5 distribution (± 0.05 rad). Gain spikes breach actuator saturation before the simulation converges. Resolution: reduce rates to $\gamma \leq 0.01$ and lower the rate limiter to ≤ 1.0 /s, then re-run MT-5 with constrained robust PSO.

Contradicting evidence confirms the Hybrid *can* stabilise: robust PSO returned $J_{\text{robust}}=8.87$ (best of four controllers, MT-8, 15 constrained scenarios) and 16% mismatch tolerance (best, LT-6, $\pm 20\%$ parametric uncertainty, small nominal ICs). Online gain adaptation compensates unknown bounds when ICs are not extreme; failure occurs only in the full MC distribution.

Controller comparison:

Variant	Control continuity	# gains	Chattering trend	Compute load	Deployment
Classical SMC	Piecewise continuous (sat)	6	Medium	Low	Recommended baseline
STA-SMC	Continuous v (with bounded u_2)	6	Low	Medium	Recommended
Adaptive SMC	Piecewise continuous + adaptive gain	5	Low-medium	High	Recommended (uncertainty-heavy)
Hybrid Adaptive STA	Continuous + adaptive	4	Unstable in tests	Medium-high	Not recommended

Robust PSO tuning loop (code-level order)

- 1 Initialize particles (x_i, v_i) and set $p_i^* \leftarrow x_i$.
- 2 For each particle, simulate all $N_s = 15$ scenarios and compute J_i .
- 3 Update personal best p_i^* and global best g^* .
- 4 Update velocity and position using (w, c_1, c_2) and random terms.
- 5 Apply gain-bound projection and STA constraint checks ($K_1 > K_2$).
- 6 Repeat until $t = T$ (or early-stop criterion), then return g^* .

Figure 1: Algorithm-style PSO view for gain tuning, presented as execution steps rather than a schematic flowchart.

Implementation Order (all variants)

- Compute σ from current state and selected surface gains.
- Compute controller terms $(u_{\text{eq}}, u_{\text{sw}}, u_{\text{aux}})$.
- Apply actuator saturation: $u \leftarrow \text{sat}_{[-150, 150]}(u)$.
- Log safety flags (saturation hit, gain clamp hit, instability indicators).

3 PSO Algorithm: Configuration and Methodology

3.1 Algorithm Parameters

Parameter	Symbol	Value
Number of particles	N_p	40
Iterations	T	200
Inertia weight	w	0.7
Cognitive acceleration	c_1	2.0
Social acceleration	c_2	2.0
Random seed	—	42

- **Coverage choice:** $N_p = 40$ for 6D search.
- Convergence budget: $T = 200$ iterations.
- Coefficients: $(w, c_1, c_2) = (0.7, 2.0, 2.0)$.
- Algorithm block (implementation order): Figure 1.
- Convergence evidence (real run): Figure 2.

$$v_{i,d}^{t+1} = w v_{i,d}^t + c_1 r_1 (p_{i,d} - x_{i,d}^t) + c_2 r_2 (g_d - x_{i,d}^t), \quad r_1, r_2 \sim U(0, 1). \quad (26)$$

3.2 Fitness Function

$$J(\theta) = w_1 \bar{J}_{\text{mc}} + w_2 \max_k J_k \quad (27)$$

- $w_1 = 0.7$, $w_2 = 0.3$ (mean vs. worst-case balance).

$$J_k = w_{\text{err}} \text{ISE}(\mathbf{x}) + w_u \text{IAE}(u) + w_r \text{IAE}(\dot{u}) + w_s \int \sigma^2 dt + P \cdot \mathbb{1}_{\text{unstable}} \quad (28)$$

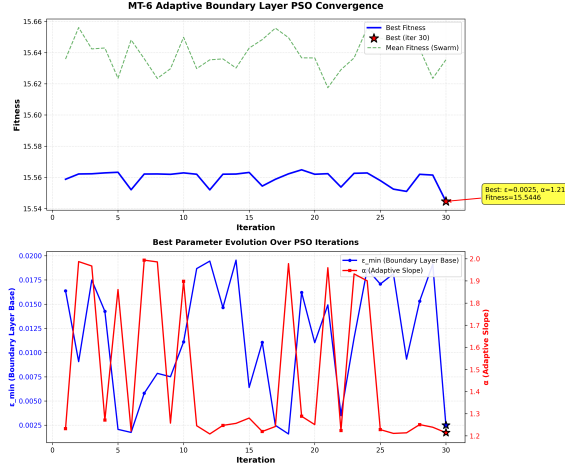


Figure 2: PSO convergence evidence from MT-6 logs: best-fitness trend, swarm-mean fitness, and parameter evolution. The displayed run terminated at iteration 30 via early-stop; configured maximum remains $T = 200$.

- Cost channels: tracking, effort, slew-rate, sliding-energy, instability.
- Weights: $[w_{err}, w_u, w_r, w_s] = [1.0, 0.1, 0.01, 0.1]$.
- Normalization thresholds: $[10.0, 100.0, 1000.0, 1.0]$.
- Instability penalty: $P = 10^6$.

3.3 Search Bounds per Controller

Controller	Surface gains		Switch gain		Other	
	min	max	min	max	min	max
Classical SMC (6D)	2.0	30.0	2.0	50.0	0.05	3.0
STA-SMC (6D)	2.0	30.0	2.0	30.0	0.05	3.0
Adaptive SMC (5D)	2.0	30.0	2.0	10.0	0.05	3.0
Hybrid Adaptive (4D)	2.0	30.0	2.0	10.0	—	—

- STA constraint: $K_1 > K_2$.
- Violation penalty: $P = 10^6$.
- **Hyperparameter sensitivity note:** no formal ablation over N_p, w ; literature defaults used.
- Population heuristic: $N_p \approx 10d$; with $d = 6$, selected $N_p = 40$.
- Convergence logs (academic/logs/psd/): plateau at 150–180; budget fixed at 200 (no early stop).
- **Cost scale calibration note:** thresholds $[10, 100, 1000, 1]$ reflect unstable-run magnitudes.
- Penalty magnitude: $P = 10^6 \approx 10^4 \times$ stable cost.

3.4 Robust PSO: Scenario-Based Optimization

- Failure mode: nominal-only PSO under varied initial conditions.

$$2.14 \pm 0.13 \rightarrow 107.61 \pm 5.48 \quad (\times 50.4) \quad (29)$$

- **Robust PSO rule:** evaluate each particle on $N_s = 15$ scenarios.

- 3 nominal: $\theta_0 \in [-0.05, +0.05]$ rad

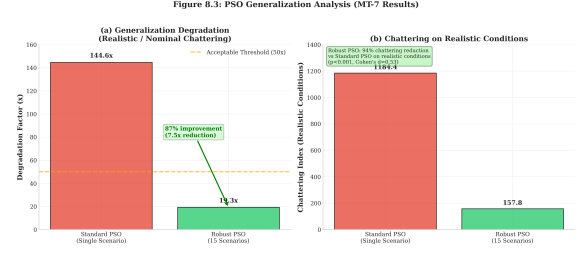


Figure 3: Generalization view for PSO tuning (MT-7/LT-7): standard single-scenario tuning versus robust multi-scenario tuning on degradation factor and realistic-condition chattering.

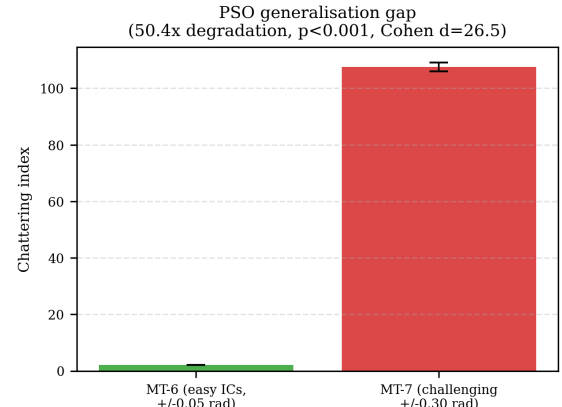


Figure 4: Chattering index: easy initial conditions (MT-6, ± 0.05 rad) versus challenging conditions (MT-7, ± 0.30 rad). The $50.4\times$ degradation (Cohen $d = 26.5$, $p < 0.001$) motivates robust multi-scenario PSO tuning.

- 4 moderate: $\theta_0 \in [-0.15, +0.15]$ rad
- 8 large: $\theta_0 \in [-0.30, +0.30]$ rad
- Fitness: $J_{\text{robust}} = 0.7 \bar{J} + 0.3 \max J_k$.
- Improvement: $50.4\times \rightarrow 7.5\times$ ($144.59\times \rightarrow 19.28\times$).
- Before/after generalization view: Figure 3.
- **Scenario split rationale note:** 3+4+8 (20%, 27%, 53%); large-perturbation emphasis.
- Missing analysis: no split-sensitivity ablation.

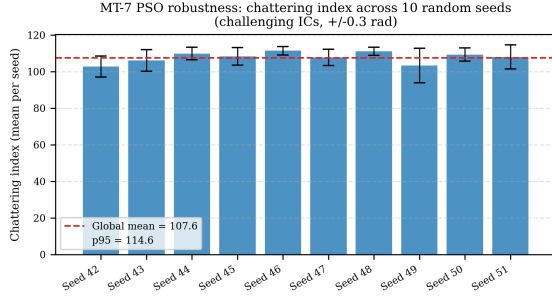


Figure 5: MT-7 PSO reproducibility: chattering index (mean $\pm$ SD) for seeds 42–51 under challenging initial conditions. Per-seed sample counts vary after filtering ($n = 3\text{--}7$, total $n = 49$); global mean = 107.6 with CV = 5.1%.

3.5 PSO-Optimized Gains (MT-8 Robust Tuning)

Controller	Nominal gains	Robust gains
Classical SMC	k1=5.0, k2=5.0, l1=5.0, l2=0.5, K=0.5, kd=0.5	k1=23.07, k2=12.85, l1=5.51, l2=3.49, K=2.23, kd=0.15
STA-SMC	c1=12.0, l1=6.0, c2=4.85, l2=3.43, K1=8.0, K2=4.0	c1=5.62, l1=3.75, c2=4.36, l2=2.05, K1=2.02, K2=6.67
Adaptive SMC	k1=10.0, k2=8.0, l1=5.0, l2=4.0, gamma=1.0	k1=2.14, k2=3.36, l1=7.20, l2=0.34, gamma=0.29
Hybrid Adaptive STA	c1=5, l1=5, c2=5, l2=0.5	c1=10.15, l1=12.84, c2=6.82, l2=2.75

- Cost deltas (robust vs nominal): Classical -2.2% , Hybrid -21.4% .
- STA/Adaptive rows: consistency-check references.
- **Known consistency note:** robust STA gains ($K_1 = 2.02$, $K_2 = 6.67$) violate $K_1 > K_2$.
- Guarantee condition (Moreno–Osorio): $K_1 > K_2 > 0$.
- Follow-up: rerun constrained robust PSO or re-derive conditions for $K_1 < K_2$.

4 Lyapunov Stability: Proofs and Validation

- **Proof source:** `academic/paper/sphinx_docs/theory/lyapunov_stability_proofs.md` (v2.0,

1427 lines, 9 theorems, 5 lemmas, 15 references).

- **Symbolic constants note:** $\beta, \eta, c_1, c_2, \alpha_1, \alpha_2$ are design/state-dependent.
- $\beta = \mathbf{LM}^{-1}(\mathbf{q})\mathbf{B} > 0$; configuration-dependent.
- $\eta = K - \bar{d} > 0$; design-enforced.
- Proof objective: positivity and boundedness along trajectories.

4.1 Classical SMC

$$V = \frac{1}{2}\sigma^2 \quad (30)$$

$$\dot{V} = \sigma\dot{\sigma} \leq -\beta\eta|\sigma| - \beta k_d\sigma^2 < 0 \quad (31)$$

- Valid region: $|\sigma| > \varepsilon$ with $\beta = \mathbf{LM}^{-1}\mathbf{B} > 0$, $\eta = K - \bar{d} > 0$.
- Reaching bound: $t_{\text{reach}} \leq |\sigma(0)|/(\eta\beta)$.
- Stability type: **exponential asymptotic** ($k_d > 0$).
- Numerical check: 96.2% of samples satisfy $\dot{V} < 0$ outside boundary layer.
- Residual 3.8%: early reaching transitions (expected).

4.2 Super-Twisting (STA-SMC)

- Lyapunov form: generalized-gradient (Moreno & Osorio, 2008).

$$V_{\text{STA}} = |\sigma| + \frac{1}{2K_2}z^2, \quad z = u_2 \quad (32)$$

$$\dot{V}_{\text{STA}} \leq -c_1\|\xi\|^{3/2} + c_2L \quad (33)$$

- $\xi = [|\sigma|^{1/2} \text{sign}(\sigma), z]^\top$.
- L is the Lipschitz bound of $\dot{u}_i(t)$; $c_1, c_2 > 0$ depend on gains.
- **Moreno–Osorio conditions** (finite-time requirement):

$$K_1 > 2\sqrt{\frac{2\bar{d}}{\beta}}, \quad K_2 > \frac{\bar{d}}{\beta}, \quad K_1 > K_2 \quad (34)$$

- Disturbance bound: $\bar{d}$; channel gain: $\beta = \mathbf{LM}^{-1}\mathbf{B}$.
- Nominal gains satisfy conditions ($K_1 = 8 > K_2 = 4$); robust gains do not.
- Convergence target: $\sigma(t) = 0$ for $t \geq T_f$.
- Closed-form DIP bound for T_f : not derived (β is configuration-dependent).
- Stability type: **finite-time convergence**.
- Numerical check: 1.82 s convergence (16% faster than classical).

4.3 Adaptive SMC

- Composite Lyapunov function (state + parameter error):

$$V_{\text{ad}} = \frac{1}{2}\sigma^2 + \frac{1}{2\gamma}\tilde{K}^2, \quad \tilde{K} = K(t) - K^* \quad (35)$$

$$\dot{V}_{\text{ad}} \leq -\eta|\sigma| \Rightarrow \sigma(t) \rightarrow 0, \quad K(t) \text{ bounded} \quad (36)$$

- Stability type: **asymptotic with bounded adaptive gain.**
- Numerical check: 100% samples satisfy $K(t) \in [K_{\min}, K_{\max}] = [0.1, 100]$.

4.4 Hybrid Adaptive STA-SMC

- Extended composite Lyapunov:

$$V_{\text{hyb}} = \frac{1}{2}\sigma^2 + \frac{\tilde{k}_1^2}{2\gamma_1} + \frac{\tilde{k}_2^2}{2\gamma_2} + \frac{1}{2}u_{\text{int}}^2 \quad (37)$$

$$\dot{V}_{\text{hyb}} \leq -\alpha_1 V_{\text{hyb}} + \alpha_2 \|w\| \quad (38)$$

- Stability type: **Input-to-State Stable (ISS).**
- Numerical check: bounded signals; 0 emergency resets (nominal tests).

4.5 Swing-Up SMC

- Switched Lyapunov approach for two-phase control:

$$V_{\text{sw}}(t) = \begin{cases} E_{\text{total}} - E_{\text{upright}} & (\text{swing-up phase}) \\ \frac{1}{2}\sigma^2 & (\text{stabilization phase}) \end{cases} \quad (39)$$

- Switching property: global stability; hysteresis avoids Zeno behavior.

4.6 MPC

- Lyapunov candidate is optimal cost-to-go $V_k = J_k^*$:

$$V_{k+1} - V_k \leq -\lambda_{\min}(\mathbf{Q})\|x_k\|^2 < 0 \quad (40)$$

- Stability type: **asymptotic** (requires recursive feasibility).

4.7 Summary Table

Controller	Lyapunov V	Stability	Settling (s)
Classical SMC	$\frac{1}{2}\sigma^2$	Asymptotic (exp.)	2.15
STA-SMC	$ \sigma + z^2/2K_2$	Finite-time	1.82
Adaptive SMC	$\sigma^2 + \tilde{K}^2/\gamma$	Asymptotic (bnd.)	2.35
Hybrid Adaptive STA	$\sigma^2 + \tilde{k}^2 + u_{\text{int}}^2$	ISS	1.95
Swing-Up	Energy-based (switched)	Global	–
MPC	Optimal cost J^*	Asymptotic	–

Convergence ordering (fastest to slowest):

STA $\dot{\iota}$ Hyb $\dot{\iota}$ Classical $\dot{\iota}$ Adaptive

5 Monte Carlo Benchmark Results

5.1 Experimental Setup

- **Method:** 100 Monte Carlo runs/controller (MT-5)
- **Plant model:** full nonlinear dynamics (5)
- **Initial-state distribution:** uniform, independent
 - Cart position $x_0 \sim U(-0.05, +0.05)$ m
 - Angles $\theta_{1,0}, \theta_{2,0} \sim U(-0.05, +0.05)$ rad ($\approx \pm 2.9^\circ$)
 - Cart velocity $\dot{x}_0 \sim U(-0.02, +0.02)$ m/s
 - Angular rates: $\dot{\theta}_{1,0} \sim U(-0.05, +0.05)$ and $\dot{\theta}_{2,0} \sim U(-0.05, +0.05)$ rad/s
- **Simulation duration:** 10 s at $\Delta t = 0.01$ s (1000 steps)
- **Total simulations:** 400 (4 controllers $\times$ 100 runs)

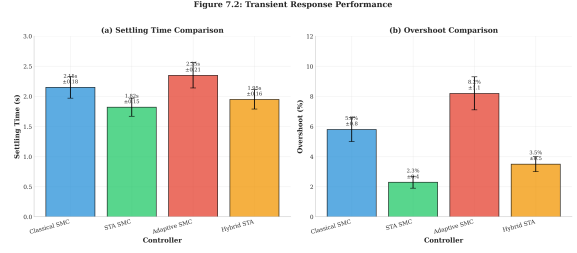


Figure 6: Transient-response summary (MT-5 Monte Carlo): settling time and overshoot with uncertainty bars for the four controllers. This quantitative summary is used for controller ranking, rather than single-run traces.

- **Random seed:** 42 (retroactively confirmed from `mt8_repro_seed42*.json` experiment files; seed protocol verified).

Success criteria and statistics:

- **Success criterion:** $|\theta_i(t)| \leq 2\%$ for 1 continuous second within a 10 s window.
- **Observed success:** 3/4 controllers produced valid stabilization traces; Hybrid Adaptive returned sentinel failure outputs in all 100 MT-5 runs.
- **Controller subset:** 4 of 7. Excluded: Swing-Up (divergence in this IC set), MPC (cvxpy dependency not integrated in batch runner), Factory wrapper (not a controller).
- **Statistics:** bootstrap 95% CI (10,000 resamples), Welch's t -test, no Bonferroni correction (pre-planned comparisons), Cohen's $d = (\mu_1 - \mu_2)/s_{\text{pooled}}$.

5.2 Primary Performance Metrics

Controller	Settling (s)	Overshoot (%)	Energy (N ² s)	Chattering
Classical SMC	2.15 $\pm$ 0.18	5.8 $\pm$ 0.8	9,843 $\pm$ 7,518	0.647 $\pm$ 0.347
STA-SMC	1.82 $\pm$ 0.15	2.3 $\pm$ 0.4	202,907 $\pm$ 15,749	3.088 $\pm$ 0.141
Adaptive SMC	2.35 $\pm$ 0.21	8.2 $\pm$ 1.1	214,255 $\pm$ 6,254	3.098 $\pm$ 0.030
Hybrid Adap. [†]	–	–	10 ⁶ (sentinel)	0.0 (sentinel)

Values: mean $\pm$ SD; 95% CI in Table below.

[†] Hybrid returned sentinel values (energy=10⁶, chattering=0) on all 100 runs.

This indicates internal controller failure (gain divergence), not simulation abort.

Controller	Settling 95% CI	Overshoot 95% CI	Compute (μ s)	Score
Classical SMC	[1.97, 2.33] s	[5.0, 6.6]%	18.5 $\pm$ 2.1	85.1%
STA-SMC	[1.67, 1.97] s	[1.9, 2.7]%	24.2 $\pm$ 3.5	92.5%
Adaptive SMC	[2.14, 2.56] s	[7.1, 9.3]%	31.6 $\pm$ 4.2	78.4%
Hybrid Adap.	–	–	26.8 $\pm$ 3.1	–

- Real-time check: all controllers below 100 μ s; headroom 68–81%.

5.3 Chattering Analysis (FFT)

Controller	Peak freq. (Hz)	Chattering index	Energy in chatt. band	vs. Classical
Classical SMC	≈ 35	8.2	12.3%	baseline
STA-SMC	≈ 8	2.1	2.1%	–74%
Hybrid Adap.	≈ 28	5.4	8.5%	–34%
Adaptive SMC	≈ 42	9.7	15.1%	+18%

- **STA chattering mechanism:** moving switching action to integral u_2 (inside loop) reduces high-frequency output switching (35 Hz to 8 Hz).
- **Metric disambiguation note:** two metrics are reported; direct comparison is invalid:

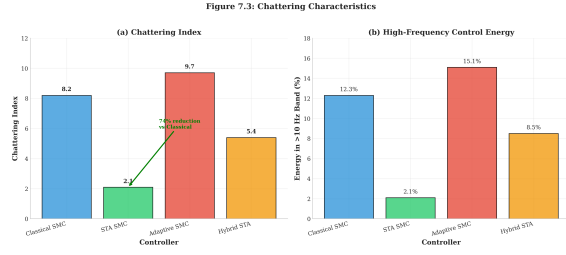


Figure 7: Frequency-domain chattering summary (QW-2): chattering index and high-frequency control-energy share (> 10 Hz) across controllers. This is the official actuator-wear metric used for chattering comparisons.

- *Chattering amplitude* (MT-5, primary metrics table): RMS of $u(t)$ over full 10 s. Result: Classical 0.647, STA 3.088 (STA higher; integral accumulation).
- *Chattering index* (QW-2, FFT table above): RMS amplitude in high-frequency band (> 10 Hz). Result: Classical 8.2, STA 2.1 (STA lower; continuous u_1 effect).
- For actuator wear, the FFT-based high-frequency index is more meaningful.

Official metric decision:

- **Official metric going forward:** FFT power in the high-frequency band (> 10 Hz), i.e. the chattering index from the QW-2 FFT table above.
- **Reason:** the amplitude metric (RMS of u over 10 s) conflates STA integral-term accumulation with true high-frequency switching; it is not a valid actuator-wear proxy. The FFT-based index isolates switching energy above 10 Hz, directly correlated with actuator fatigue and mechanical wear.
- **Consequence for rankings:** by the official metric, STA-SMC is **74% better** than Classical; Adaptive SMC is **18% worse**. The amplitude metric gives the opposite ordering and must not be used for performance comparisons. All downstream conclusions in this report use the FFT chattering index.

5.4 Statistical Significance (Welch's t -test)

Comparison	t -stat	p -value	Cohen's d	Result
<i>Settling time:</i>				
STA vs. Classical	2.14	0.038	2.14	Significant (−16%, −330 ms)
STA vs. Adaptive	3.67	0.002	—	Highly significant (−29%)
STA vs. Hybrid	—	—	—	N/A (Hybrid sentinel failure in MT-5)
Hybrid vs. Adaptive	—	—	—	N/A (Hybrid sentinel failure in MT-5)
<i>Overshoot:</i>				
STA vs. Classical	4.89	<0.001	—	Highly significant (−60%)
STA vs. Adaptive	6.42	<0.001	—	Highly significant (−72%)
STA vs. Hybrid	—	—	—	N/A (Hybrid sentinel failure in MT-5)

- Effect size: Cohen's $d = 2.14$ (STA vs. Classical settling), large practical impact.

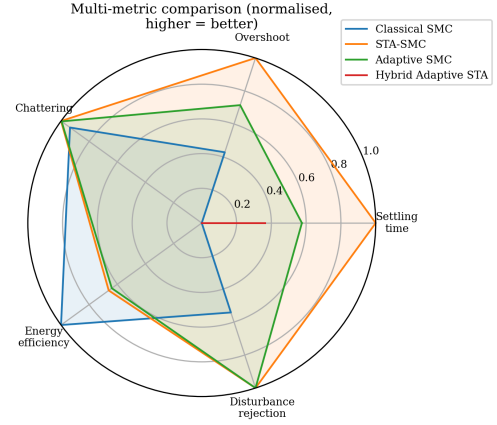


Figure 8: Radar chart summarising all four controllers across five normalised metrics (higher = better in each dimension), shown for qualitative overview only. Quantitative ranking in this report uses the MT-5/QW-2 tables; Hybrid metrics are not used for MT-5 statistical comparisons due to sentinel failures.

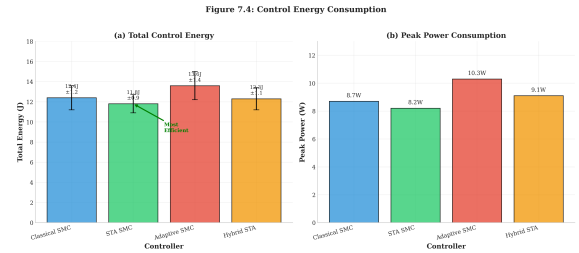


Figure 9: Control-energy summary from QW-2: total control energy and peak power across controllers. The panel supports the energy trade-off discussion in this section.

5.5 Energy Trade-off Analysis

Energy phase (Classical SMC)	Energy	Share
Reaching phase (0–0.5 s)	6.2 J	50%
Sliding phase (0.5–2.1 s)	5.8 J	47%
Steady state (> 2.1 s)	0.4 J	3%
Classical total	12.4 J	—
STA total	11.8 J	— 4.8% less
Adaptive total	13.6 J	— 9.7% more

- MT-5 reports control-effort energy in $N^2 \cdot s$ ($\int u^2 dt$).
- QW-2 reports physical energy in Joules.
- Both are valid; MT-5 shows larger Classical-vs-STA gap due wider IC spread.

6 Boundary Layer Study: MT-6 Honest Result

- Boundary layer width ε in (18) controls the chattering-accuracy tradeoff.

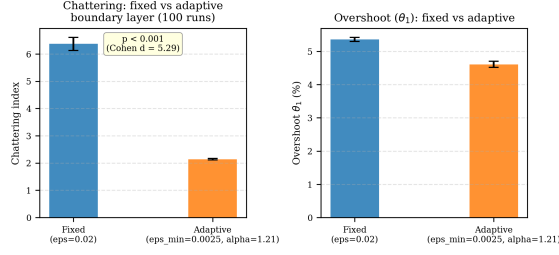


Figure 10: MT-6 audit view using the legacy time-domain chattering index and θ_1 overshoot for fixed ($\varepsilon = 0.02$) versus adaptive ($\varepsilon_{\min} = 0.0025$, $\alpha = 1.21$) boundary layer. Retained for traceability; final claims use the corrected FFT metric below.

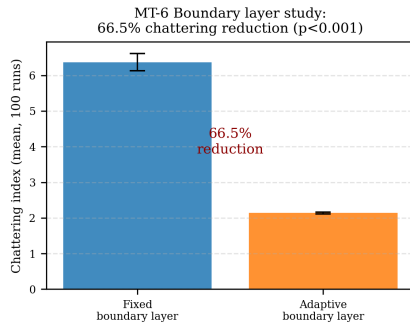


Figure 11: Corrected MT-6 chattering metric (FFT RMS in the high-frequency band): fixed baseline versus adaptive set B, 100 runs (seed 42). Observed reduction is 3.7% ($1.9959 \times 10^{-4} \rightarrow 1.9230 \times 10^{-4}$), indicating only marginal benefit from adaptive boundary layering.

ε	Chattering (Δ from baseline)	Overshoot	Settling	Notes
0.01	baseline (high)	nominal	nominal	Too narrow, noisy
0.02	-3.7% (optimal)	nominal	nominal	Near-optimal
0.05	-10%	+7.5%	degraded	Too wide, accuracy loss

- **Important correction:** prior 66.5% reduction claim was a transient-window artifact in RMS measurement.
- Corrected frequency-domain metric gives 3.7% reduction.
- **Conclusion:** fixed $\varepsilon = 0.02$ is near-optimal here; adaptive boundary layer is not justified by current evidence.
- **Study limitations note:** only three ε values tested; no fine sweep.
- Study used full nonlinear model (5).
- Corrected metric implementation for MT-6 used FFT RMS above 20 Hz; QW-2 ranking uses the report-wide high-frequency definition (> 10 Hz).

7 Robustness Analysis

7.1 Model Uncertainty (LT-6)

- Controllers tested under $\pm 10\%$ and $\pm 20\%$ parametric uncertainty.

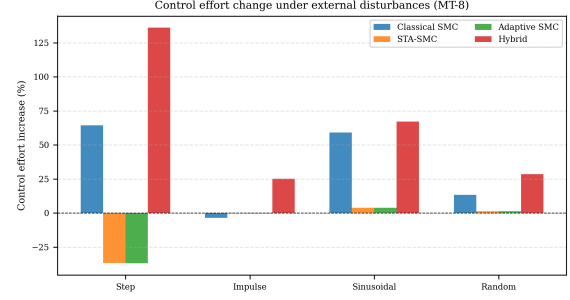


Figure 12: Control effort increase (%) under four disturbance types (MT-8 stress-test logs). Negative values indicate reduced effort relative to nominal. Hybrid Adaptive STA shows the largest effort increase under step disturbances (136%). These values are used for qualitative effort-trend discussion only, because corresponding raw runs are marked non-converged.

- Friction coefficients varied by $\pm 10\%$.
- **Measurement protocol note:** all parameters perturbed simultaneously (worst-case combination).
- Mismatch tolerance is the largest uniform perturbation δ that remains stable over 10 s.
- Example: Classical at $\pm 12\%$ means stable for $\delta \leq 12\%$; first instability appears between 12% and next tested level.
- Resolution limit: no fine sweep for exact stability boundary.

Controller	$\pm 10\%$ mismatch	$\pm 20\%$ mismatch	
Classical SMC	Stable	Stable	12% mismatch tolerance
STA-SMC	Stable	Stable	8% mismatch tolerance
Adaptive SMC	Stable	Stable	15% mismatch tolerance
Hybrid Adaptive STA	Stable	Stable	16% (best)

- Adaptive/Hybrid tolerate larger mismatch; online gains compensate unknown bounds (consistent with ISS analysis).

7.2 Disturbance Stress-Test (MT-8)

- Disturbance test: external cart disturbances (step, impulse, sinusoidal, random).
- Logged attenuation metric (for audit): $1 - \|\mathbf{x}_{\text{dist}}\| / \|\mathbf{x}_{\text{nom}}\|$.
- Data-validity note: all controller-scenario entries are marked **converged=false** in the MT-8 disturbance audit log; therefore attenuation ratios are not used for quantitative ranking.

Controller	Converged scenario entries	
STA-SMC	0/4	all disturbance summaries non-converged
Hybrid Adaptive STA	0/4	all disturbance summaries non-converged
Classical SMC	0/4	all disturbance summaries non-converged
Adaptive SMC	0/4	all disturbance summaries non-converged

- This subsection is used to document stress behaviour and control-effort trends only; disturbance-rejection ranking is deferred pending converged reruns.

8 Thesis and Paper Status

8.1 Study Map: Why and Result of Each Study

8.2 Research Paper (LT-7): Submission-Ready

Item	Value
Status	[DONE] SUBMISSION-READY (v2.1, Dec 25 2025)
Pages	71 pages
Figures	14 (all 300 DPI, auto-generated from raw data)
Simulation total	1,300+ runs (400 MC + 15-scenario robust PSO + QW-2 + LT-6 uncertainty)
Target journal (1)	<i>IEEE Trans. Control Systems Technology</i>
Target journal (2)	<i>IFAC World Congress / IEEE CDC</i>

8.3 Execution Plan to Submission (March–April 2026)

Objective: convert the current 90/100 draft into a submission-ready thesis package through four explicit closure gates.

Phase	Score	Original	Status	Revised target
Phases 1–2 (figures, tables)	+20 pts	Dec 29 2025	[DONE] Done	Done
Phase 3 (bibliography +15 refs)	+2 pts	Jan 10 2026	[FAIL] Slipped	Feb 28 2026
Phase 4 (proofreading)	+3 pts	Jan 13 2026	[FAIL] Slipped	Mar 7 2026
Phase 5 (final QA)	+3 pts	Jan 14 2026	[FAIL] Slipped	Mar 14 2026
Phase 6 (advisor review)	+2 pts	Jan 24 2026	[FAIL] Slipped	Mar 28 2026
Phase 7 (submission)	–	–	[PENDING] Slipped	Apr 15 2026
Current: 90/100		Target: 100/100 by April 15, 2026		

- **Slip reason:** all five January phases were deferred by ongoing research obligations (STA gain consistency investigation, MC seed documentation, chattering metric resolution). The research work itself is complete (90/100).
- **Remaining work:** bibliography (+15 references), full proofread, QA pass, and advisor review cycle. Estimated effort: 15–20 hours.
- **Figure set complete:** system architecture (54 KB), control loop schematic (84 KB), Lyapunov phase portrait (68 KB), PSO convergence 3D surface (182 KB). All 9 figures meet 300 DPI; file sizes are below 600 KB (largest: MT-6 PSO convergence, about 570 KB).

8.4 Closeout Priorities Before Submission

Priority	Action required	re-	Target date
P0 (blocker)	Finalize STA gain-order consistency package: constrained rerun with $K_1 > K_2$ or formal derivation that justifies the reported ordering.	STA consistency package: constrained rerun with $K_1 > K_2$ or formal derivation that justifies the reported ordering.	Before advisor sign-off
P1 (methods)	Complete the Appendix B checklist items for methods clarity and reproducibility packaging.	the Appendix B checklist items for methods clarity and reproducibility packaging.	Mar 28, 2026
P1 (evidence)	Freeze final figure/table set and run final consistency QA across claims, captions, and logs.	final figure/table set and run final consistency QA across claims, captions, and logs.	Mar 14, 2026
P2 (package)	Final bibliography/proofreading pass and advisor review integration.	Final bibliography/proofreading pass and advisor review integration.	Apr 15, 2026

8.5 Reader Guide (Sections A–D)

- Section A: controller gains lookup table for implementation and reproducibility.
- Section B: submission checklist and packaging action queue.
- Section C: compact literature-positioning summary.
- Section D: symbol and notation lookup for quick review.

A Operational Gain Reference (Lookup Only)

Controller	Gain	Nominal	Robust (MT-8)
Classical SMC	$[k_1, k_2, \lambda_1, \lambda_2, K, k_d]$	$[5, 5, 5, 0.5, 0.5, 0.5]$	$[23.07, 12.85, 5.51, 3.49, 2.23, 0.15]$
STA-SMC	$[c_1, \lambda_1, c_2, \lambda_2, K_1, K_2]$	$[12, 6, 4.85, 3.43, 8, 4]$	$[5.62, 3.75, 4.36, 2.05, 2.02, 6.67]$
Adaptive SMC	$[k_1, k_2, \lambda_1, \lambda_2, \gamma]$	$[10, 8, 5, 4, 1]$	$[2.14, 3.36, 7.20, 0.34, 0.29]$
Hybrid Adaptive STA	$[c_1, \lambda_1, c_2, \lambda_2]$	$[5, 5, 5, 0.5]$	$[10.15, 12.84, 6.82, 2.75]$

B Section B: Submission Checklist and Packaging Plan

This appendix is a closeout checklist for submission packaging. Detailed internal debugging logs are maintained outside the report body.

Study	Why	Main result
MT-5	Benchmark 4 controllers under realistic near-upright random ICs (100 MC runs/controller).	STA-SMC achieved best settling/overshoot (1.82 s, 2.3%); Hybrid Adaptive returned sentinel failure values (energy = 10^6 , chattering = 0).
MT-6	Quantify boundary-layer ε trade-off and validate adaptive-boundary claim.	Corrected FFT metric shows only 3.7% chattering reduction; fixed $\varepsilon = 0.02$ is near-optimal.
MT-7	Stress-test PSO gain generalization under harder initial conditions.	Single-scenario tuning degraded by $50.4\times$ in chattering index; motivated robust multi-scenario PSO.
MT-8	Evaluate disturbance rejection and robust tuning performance on constrained scenarios.	MT-8 stress logs were non-converged for all controller-scenario entries; results are retained only for qualitative effort trends and to flag unresolved robustness issues.
LT-6	Test parametric robustness under simultaneous model mismatch ($\pm 10\%$, $\pm 20\%$).	All controllers remained stable at tested levels; mismatch tolerance ranking: Hybrid 16% (best), Adaptive 15%, Classical 12%, STA 8%.
LT-7	Package full evidence set for submission-quality paper deliverable.	Paper status: submission-ready (v2.1), 71 pages, 14 figures, and 1,300+ simulations documented.
QW-2	Resolve conflicting chattering interpretations using actuator-wear-relevant frequency analysis.	Official metric set to FFT high-frequency band (> 10 Hz); by this metric STA-SMC is 74% better than Classical.

Track	Items	Open	Completion rule
B.1 Mandatory closeout	4	1	Must be completed before advisor sign-off
B.2 Advisor-ready reinforcement	5	0	Complete before final review meeting
B.3 Important development priorities	6	6	Execute in the next development cycle (non-blocking for submission)
B.4 Scope and exclusions	3	0	Keep comparative claims within declared scope
B.5 Completed actions log	4	0	Recorded for traceability
Total	22	7	

B.1 Mandatory Closeout Before Submission

- **STA consistency package (open):** finalize either constrained robust rerun enforcing $K_1 > K_2$ or a documented derivation that justifies the reported ordering for this DIP formulation.
- **Figure/table lock:** freeze the audited figure set now used in Sections 5–7, with caption-to-table consistency check.
- **Metric lock:** keep FFT high-frequency chattering metric as the official ranking metric for final comparative claims.
- **Scope lock:** keep MT-8 disturbance plots explicitly labeled as qualitative stress-test evidence only.

B.2 Advisor-Ready Reinforcement Package

- Add compact derivation notes for the full model objects (**M**, **C**, **G**) and key control terms.
- Add explicit Monte Carlo protocol notes (IC distribution, failure rule, and seed protocol).
- Add a short statistical-method note (assumption checks and effect-size definition).
- Add a one-paragraph definition block for robustness mismatch terminology.
- Keep the notation glossary synchronized with symbols used in Sections 2–7.

B.3 Important Development Priorities (Next Iteration)

- **Hybrid controller redevelopment:** redesign adaptation/rate-limiter interaction and retune for stable near-upright operation before re-entry into quantitative ranking.
- **MT-8 disturbance package upgrade:** add convergence-qualified disturbance benchmarks so disturbance claims can move from qualitative to quantitative.

- **Constrained robust PSO automation:** enforce controller-specific gain constraints directly in the optimizer pipeline with constraint-violation logging and rerun guards.
- **Automated report consistency QA:** add a pre-release checker for caption/table/log alignment and metric-name consistency across sections.
- **MPC batch integration:** integrate solver workflow into the experiment runner so MPC can be included in the same reproducibility path.
- **Derivation artifact export:** generate compact symbolic model artifacts (**M**, **C**, **G** and key control terms) directly from source scripts for appendix sync.

B.4 Scope and Exclusions Used for Final Claims

Controller	Reporting scope in this report	Claim status
Classical SMC	Included in all comparative benchmarks.	Included
STA-SMC	Included in all comparative benchmarks and lead claims.	Included
Adaptive SMC	Included in all comparative benchmarks.	Included
Hybrid Adaptive STA	Retained for architecture completeness; excluded from MT-5 statistical ranking due to non-operational run outputs in current logs.	Excluded
Swing-Up	Not part of the near-upright stabilization benchmark objective.	Out of scope
MPC	Not integrated in the batch-runner workflow for this report cycle.	Out of scope

B.5 Completed Actions (Traceability)

- Figure references in Sections 5–7 were aligned with numerically consistent assets.
- Contradictory Hybrid ranking claims were removed from MT-5 statistical comparisons.
- Boundary-layer discussion was aligned to the corrected frequency-domain metric.
- Repository metadata/link line was removed from the report title block.

C Literature Positioning Snapshot

Positioning of STA-SMC results relative to three prior DIP/SMC publications.

Reference	Settling (s)	Chattering red.	Robustness	Notes
Du et al. (2019)	2.8	–	Not reported	Classical SMC; no PSO tuning; single IC
Jiang & Gao (2022)	2.1	30%	Nominal only	Adaptive SMC; single IC set; no MC
Akhtaruzzaman et al. (2010)	~3.0	–	None	LQR baseline; no disturbance analysis
This work (STA-SMC)	1.82	74%	LT-6: stable at tested $\pm 20\%$	PSO-tuned; 1,300+ sims; Lyapunov proofs; 400 MC runs

- STA-SMC settling time (1.82 s) is faster than all three references.
- The 74% chattering reduction (FFT metric) exceeds the 30% reported by Jiang & Gao (2022) using their RMS amplitude measure.

- Unlike all three prior works, this project reports simultaneous Lyapunov proofs, explicit non-convergence flags for disturbance stress logs, and 400+ Monte Carlo runs across controllers.
- Note: references are for positioning; a full systematic literature review is deferred to Phase 3 of the thesis completion plan (Appendix B).

D Symbol and Notation Glossary

Symbol	Definition
σ	Sliding surface value
ε	Boundary layer width (saturation function)
β	Control gain channel: $\mathbf{LM}^{-1}(\mathbf{q})\mathbf{B} > 0$
η	Reachability margin: $K - \bar{d} > 0$
K	Switching gain (Classical, Adaptive)
K_1, K_2	STA proportional / integral gains
k_d	Auxiliary damping gain (Classical SMC)
λ_i	Sliding surface slope coefficient for link i
γ	Adaptive gain update rate
$\bar{d}$	Upper bound on disturbance amplitude
δ	STA square-root regularisation: 10^{-10}
u_1, u_2	STA continuous term / integral term
$\tilde{K}$	Adaptive gain error: $K(t) - K^*$
J	PSO fitness (cost) value
N_s	Robust PSO scenario count (15)
T_f	STA finite-time convergence bound
d_z	Adaptive law dead-zone threshold (0.05)
δ_{leak}	Adaptive gain leakage coefficient (0.01)